

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
Last login: Mon Aug  3 11:12:29 on ttys000
varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_LAB
varnikakomali@VARNIKAs-MacBook-Air OS_LAB % cd Practical
varnikakomali@VARNIKAs-MacBook-Air Practical % nano question1.c
varnikakomali@VARNIKAs-MacBook-Air Practical % gcc question1.c -o question1
varnikakomali@VARNIKAs-MacBook-Air Practical % ./question1
Enter a command (e.g., ls, date, pwd): uname

[Parent Process] PID: 19735 | Created Child PID: 19744
[Parent Process] Waiting for child to finish...

[Child Process]  PID: 19744 | Parent PID: 19735
[Child Process]  Executing command: uname
Darwin
[Parent Process] Child process completed. Exiting.
varnikakomali@VARNIKAs-MacBook-Air Practical %
```

```
varnikakomali@VARNIKAs-MacBook-Air Practical % sysctl -a | grep machdep.cpu
machdep.cpu.cores_per_package: 8
machdep.cpu.core_count: 8
machdep.cpu.logical_per_package: 8
machdep.cpu.thread_count: 8
machdep.cpu.brand_string: Apple M2
varnikakomali@VARNIKAs-MacBook-Air Practical %
```

```
varnikakomali@VARNIKAs-MacBook-Air Practical % diskutil list
/dev/disk0 (internal, physical):
#:

| #: | TYPE                  | NAME            | SIZE      | IDENTIFIER |
|----|-----------------------|-----------------|-----------|------------|
| 0: | GUID_partition_scheme |                 | *251.0 GB | disk0      |
| 1: | Apple_APFS_ISC        | Container disk1 | 524.3 MB  | disk0s1    |
| 2: | Apple_APFS            | Container disk3 | 245.1 GB  | disk0s2    |
| 3: | Apple_APFS_Recovery   | Container disk2 | 5.4 GB    | disk0s3    |


/dev/disk3 (synthesized):
#:

| #: | TYPE                  | NAME                    | SIZE      | IDENTIFIER |
|----|-----------------------|-------------------------|-----------|------------|
| 0: | APFS Container Scheme | -                       | +245.1 GB | disk3      |
|    |                       | Physical Store disk0s2  |           |            |
| 1: | APFS Volume           | Macintosh HD            | 13.5 GB   | disk3s1    |
| 2: | APFS Snapshot         | com.apple.os.update-... | 13.5 GB   | disk3s1s1  |
| 3: | APFS Volume           | Preboot                 | 17.9 GB   | disk3s2    |
| 4: | APFS Volume           | Recovery                | 2.6 GB    | disk3s3    |
| 5: | APFS Volume           | Data                    | 144.6 GB  | disk3s5    |
| 6: | APFS Volume           | VM                      | 3.2 GB    | disk3s6    |


/dev/disk5 (disk image):
#:

| #: | TYPE                  | NAME                    | SIZE      | IDENTIFIER |
|----|-----------------------|-------------------------|-----------|------------|
| 0: | GUID_partition_scheme |                         | +692.4 MB | disk5      |
| 1: | Apple_HFS             | mysql-8.0.46-macos15... | 692.3 MB  | disk5s1    |


varnikakomali@VARNIKAs-MacBook-Air Practical %
```

```
[varnikakomali@VARNIKAs-MacBook-Air Practical % ps
PID TTY          TIME CMD
19724 ttys000      0:00.04 -zsh
varnikakomali@VARNIKAs-MacBook-Air Practical %
```

```
Processes: 904 total, 2 running, 902 sleeping, 3606 threads
Load Avg: 1.82, 1.98, 2.01  CPU usage: 6.26% user, 7.27% sys, 86.46% idle
SharedLibs: 1040M resident, 194M data, 152M linkedit.
MemRegions: 0 total, 0B resident, 506M private, 2254M shared.
PhysMem: 15G used (2009M wired, 1494M compressor), 281M unused.
VM: 385T vsize, 6144M framework vsize, 308301(0) swapins, 466182(0) swapouts.
Networks: packets: 152107604/98G in, 29027354/13G out. Disks: 22343422/575G read, 36398913/588G written.
```

PID	COMMAND	%CPU	TIME	#TH	#WQ	#PORT	MEM	PURG	CMPRS	PGRP	PPID	STATE	BOOSTS
596	WindowServer	33.3	40:23:36	22	6	9324+	514M-	55M-	134M	596	1	sleeping	*0[1]
0	kernel_task	25.7	19:03:20	577/9	0	0	54M-	0B	0B	0	0	running	0[0]
19766	top	10.1	00:01.11	1/1	0	29	9248K	0B	0B	19766	19724	running	*0[1]
19767	screencaptur	7.1	00:00.20	4	3	99	7664K+	752K	0B	1274	1274	sleeping	*1[2]
1225	duetexpertd	3.3	02:38:26	8	7	458	21M+	1408K	3424K	1225	1	sleeping	0[53210]
9787	Google Chrom	2.9	20:39:53	17	1	167	82M-	0B	22M	9780	9780	sleeping	*1635+[5118]
676	com.apple.Dr	2.7	02:55:52	6	4	1393	30M	0B	2912K	676	1	sleeping	0[1]
9780	Google Chrom	2.5	44:19.44	51	3	1547-	346M-	528K	80M	9780	1	sleeping	*4717+[14161]
1093	WindowManage	2.0	28:26.47	6	3	709	18M	0B	5296K	1093	1	sleeping	*0[326472+]
636	mDNSResponde	1.6	52:55.13	3	1	107	9041K	0B	864K	636	1	sleeping	*0[1]
9786	Google Chrom	1.5	67:15.54	23	3	431	421M-	137M+	39M	9780	9780	sleeping	*1[165428]
41260	com.apple.Vi	1.3	89:49.19	22	1	196	1172M	0B	1503M	41260	1	sleeping	*27115[3]
953	BiomeAgent	1.2	16:05.20	3	2	315-	22M-	384K	15M	953	1	sleeping	0[125094]
19220	Google Chrom	0.8	00:50.86	31	1	399	203M+	240K	0B	9780	9780	sleeping	*3[6694+]
605	runningboard	0.5	96:06.53	8	7	1378	12M	0B	992K	605	1	sleeping	*4[1]
19768	screencaptur	0.5	00:00.10	6	4	192	8913K	0B	0B	19768	1	sleeping	*0[42+]
955	linkd	0.4	01:28.18	4	3	1069+	13M+	0B	7520K	955	1	sleeping	0[5174]
19619	Terminal	0.4	00:08.60	11	4	435	75M+	33M	0B	19619	1	sleeping	*0[238+]
97555	mysqld	0.4	09:39.61	42	0	65	418M	0B	416M	97555	1	sleeping	*0[1]
936	cfprefsd	0.2	09:02.36	3	2	1296	4720K	416K	1024K	936	1	sleeping	*510[1294]
527	fseventsd	0.2	18:17.93	13	1	198	4592K	0B	992K	527	1	sleeping	*0[1]
523	logd	0.2	37:06.16	4	3	2970	13M	0B	14M	523	1	sleeping	*0[1]
537	powerd	0.1	12:38.00	3	2	162	5632K	0B	656K	537	1	sleeping	*0[1]
591	corebrightne	0.1	53:26.53	2	1	146	4656K	0B	1360K	591	1	sleeping	*0[1]
19295	netbiosd	0.1	00:10.34	5	4	29	2416K	0B	0B	19295	1	sleeping	*0[1]
571	locationd	0.1	44:39.68	6	3	312	12M	256K	2176K	571	1	sleeping	0[274260]
655	audioclocks	0.0	10:34.94	3	2	53	2912K	0B	1200K	655	1	sleeping	*0[1]
628	airportd	0.0	02:36:10	9	7	395	18M	0B	1936K	628	1	sleeping	*1[1]
1098	gamepolicyd	0.0	04:45.45	4	3	93+	5440K	0B	1088K	1098	1	sleeping	*2+[105083+]
820	mds_stores	0.0	38:04.76	6	4	157-	56M-	0B	33M	820	1	sleeping	*0[1]
552	mds	0.0	32:59.69	9	6	533	72M	0B	54M	552	1	sleeping	*0[1]
19379	Google Chrom	0.0	02:56.46	23	1	309	232M	0B	0B	9780	9780	sleeping	0[71238]
566	launchservic	0.0	06:51.06	5	4	2974	14M	0B	6720K	566	1	sleeping	*30466[221]
947	universalacc	0.0	01:29.84	9	4	261	9121K	0B	3456K	947	1	sleeping	*0[157081+]